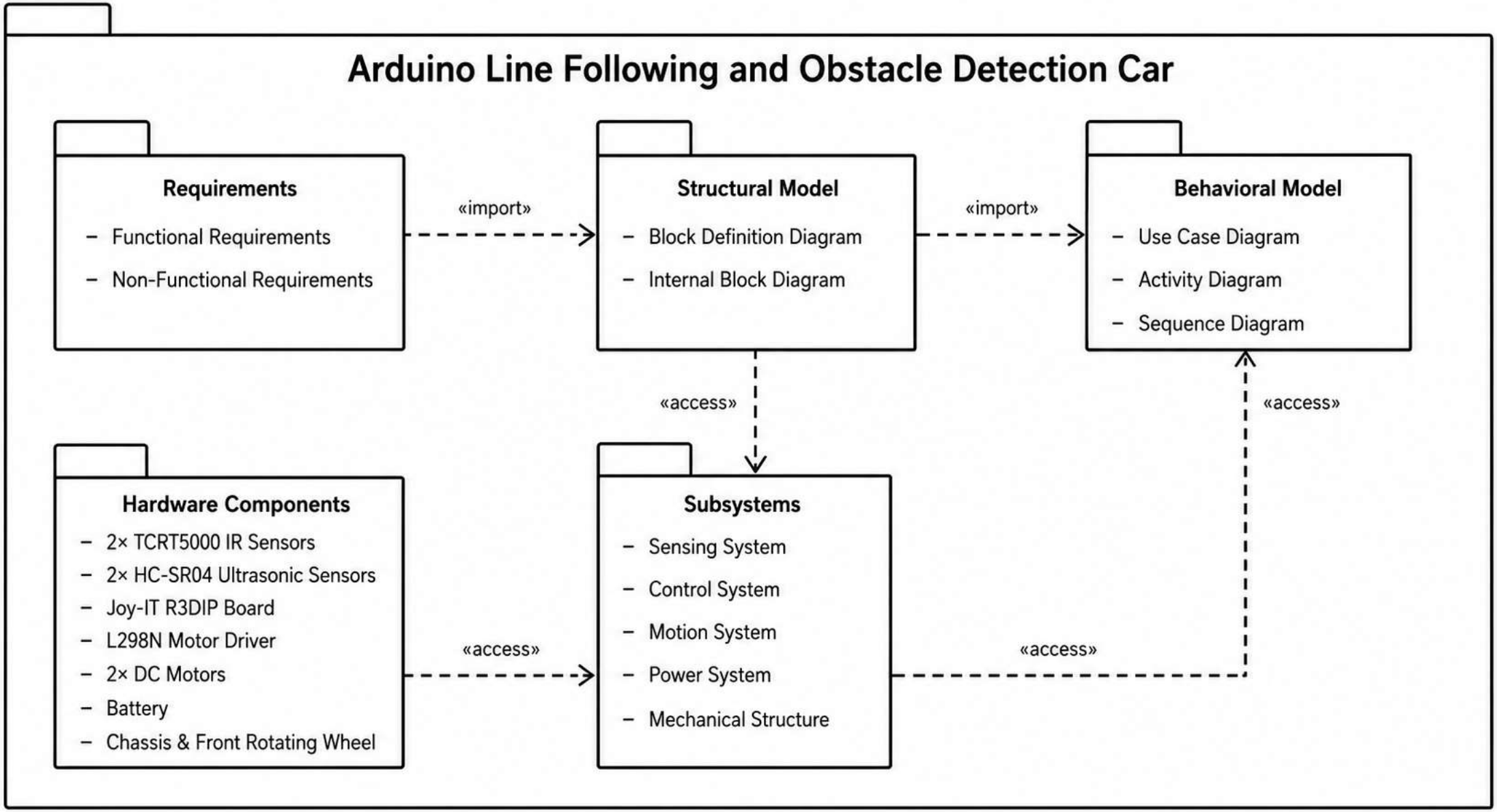

Arduino Line Following and Obstacle Detection Car
Requirements
– Functional Requirements
– Non-Functional Requirements
«import»
Structural Model
– Block Definition Diagram
– Internal Block Diagram
«import»
Behavioral Model
– Use Case Diagram
– Activity Diagram
– Sequence Diagram
«access»
«access»
Hardware Components
– 2× TCRT5000 IR Sensors
– 2× HC-SR04 Ultrasonic Sensors
– Joy-IT R3DIP Board
– L298N Motor Driver
– 2× DC Motors
– Battery
– Chassis & Front Rotating Wheel
«access»
Subsystems
– Sensing System
– Control System
– Motion System
– Power System
– Mechanical Structure
«access»